

Cabling and Connections Guide for Model 4225-PMU Ultra Fast I-V Module and Model 4225-RPM Remote Preamplifier/Switch

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INTRODUCTION

The Model 4225-PMU Ultra Fast I-V Module is in a new class of pulse instruments that output pulses and simultaneously takes high speed measurements. In particular, the Model 4225-PMU enables the user to make I-V measurements with voltage pulse widths down to 70 ns with 5 ns sampling intervals for both current and voltage measurements. In this higher frequency range, it is important to use the proper cabling and connections to avoid incorrect results and permit the shortest possible pulse widths. Improper connections can cause the pulse shape to be distorted as well as affect the power transmitted to the device. This application note explains how to make proper connections to the device under test for optimal results and then how to implement the appropriate connections to the Model 4225-PMU and the Model 4220-PGU Pulse Generator Unit. The Model 4220-PGU is a pulse-only card which has the same pulse output and timing specs as the 4225-PMU, but has no measurement capabilities. This application note also provides detailed information on how to make appropriate connections to the Model 4225-RPM Remote Amplifier/Switch option for the Model 4225-PMU.

MODEL 4225-PMU AND MODEL 4220-PGU CONNECTIONS

The Model 4225-PMU replaces traditional pulse/measure hardware configurations, which typically include a pulse generator, a multi-channel oscilloscope, and specially-designed interconnect hardware like bias tees and various adaptors. The set-up and connections of these systems were complicated. The Model 4225-PMU was designed to make easy connections to the DUT. This section will illustrate how to make connections from the PMU to two, three, and four terminal devices. Note that the 4225-PMU and 4220-PGU have two independent, identical channels, so the connections illustrated could be interchanged between Channel 1 (CH1) and Channel 2 (CH2).

PMU and PGU Connections:

Each Model 4225-PMU and 4220-PGU has two identical channels. Each channel is terminated with female SMA. Drawings of the 4225-PMU and 4220-PGU are shown in Figure 1. Since the connections for PMU and PGU are the same (except for the RPM connections), this app note will now only refer to the PMU for simplicity.

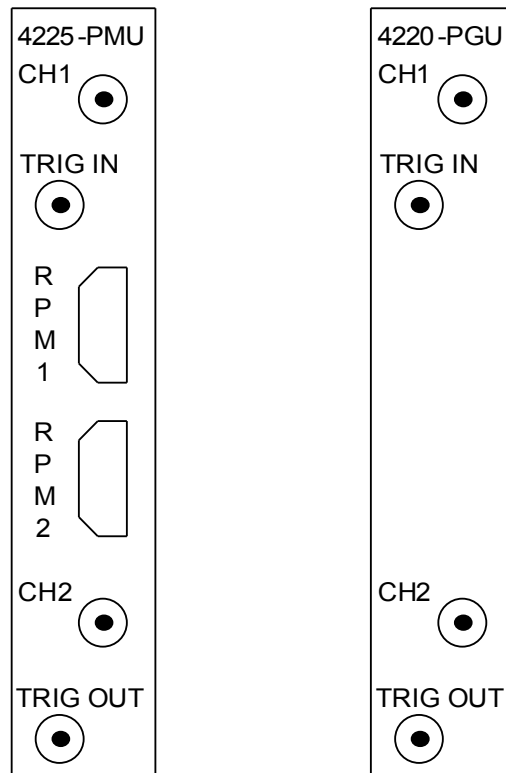


Figure 1. Front Panel Drawings of Model 4225-PMU and 4220-PGU

A simple schematic of the 4225-PMU showing the coax cables connected to the output of each channel is shown in Figure 2.

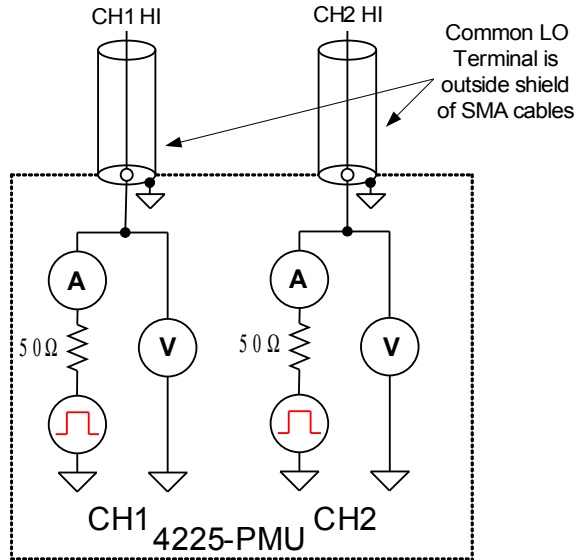


Figure 2. Simplified schematic of Model 4225-PMU with two coax cables

The center conductor of each SMA cable is the HI terminal of the PMU channel. The outside shields of the coax cables is the common LO terminal of the PMU.

Two-terminal devices:

The Model 4225-PMU enables the user to make ultra fast I-V measurements on two terminal devices using either one or two channels of the PMU. Figure 3 shows the connections to a resistor using a single channel. Notice one end of the resistor is connected to the center conductor of the coax cable (HI) of CH1 and the other side of the resistor is connected to PMU common (LO), or the outside shield of the coax cable.

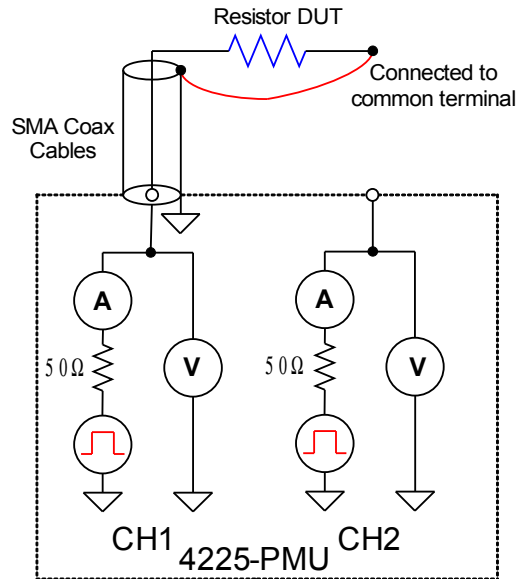


Figure 3. Connecting one channel of the PMU to a two terminal device

A two terminal device can also be connected using two channels of the PMU as shown in Figure 4. The HI terminal of CH1 is connected to one end of the device and the HI terminal of CH2 is connected to the other end of the device. In some applications, like testing a high resistance device, it is advantageous to have one channel pulse voltage and the other channel measure the resulting current. Note the shields of the coax cables are connected together close to the device under test.

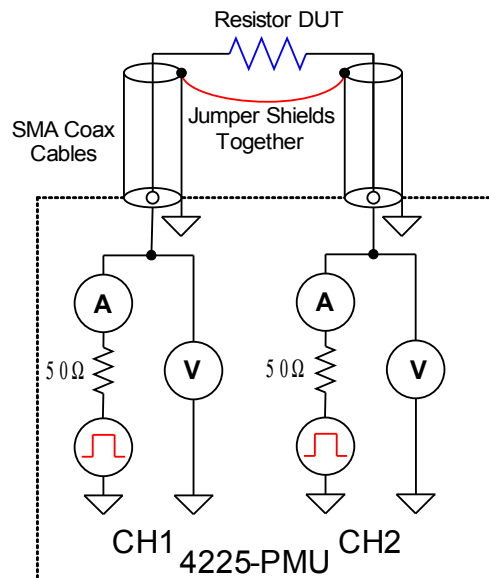


Figure 4. Using both channels of the PMU to connect to a two terminal device

Three Terminal Devices:

A three terminal device can be connected using either two or three PMU channels depending on if sourcing and/or measuring need to be performed at each device pin. An example of both channels of a single PMU connected to a 3 terminal MOSFET is shown in Figure 5. In this diagram, the Gate terminal is connected to CH1 and the Drain terminal is connected to CH 2. The Source terminal is connected to the outside shield, or common terminal, of the PMU.

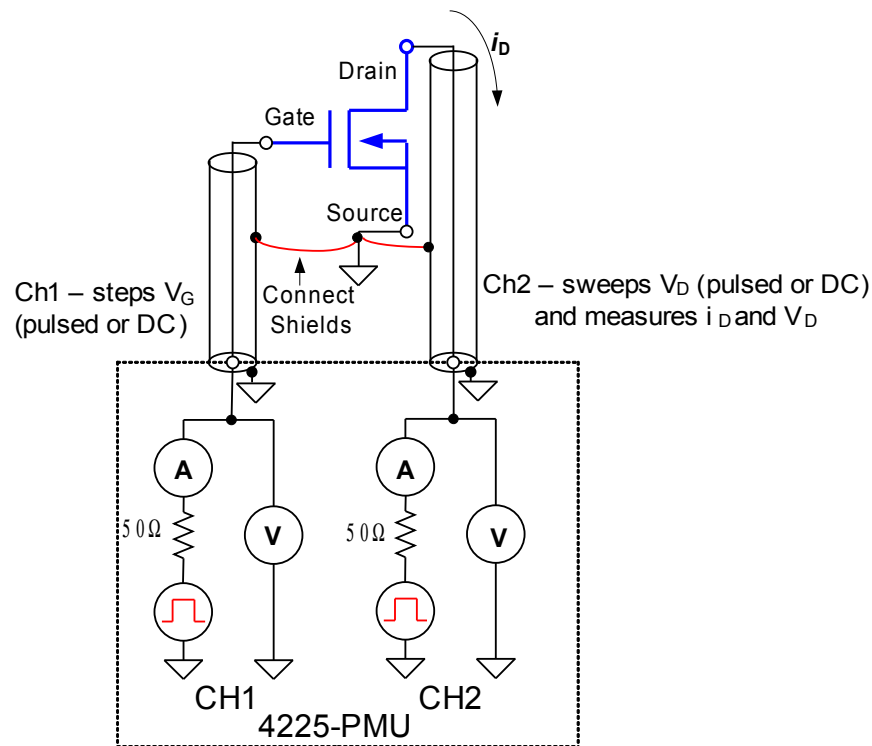


Figure 5. Connections from the two channels of the PMU to a three terminal MOSFET

If ultra fast I-V sourcing and measuring is necessary at each device terminal, then a second PMU is required for the three terminal device. Up to four PMUs (eight channels) can be installed in one 4200-SCS mainframe.

Four Terminal Devices:

To test a four terminal device, two PMUs are usually required. Figure 6 shows the four PMU channels connected to a four-terminal MOSFET. This configuration enables the

user to have complete flexibility of pulsing and measuring at any terminal on the device, similar to using 4 SMUs to characterize a 4-terminal device. Notice the shields of the coax cable from all four channels are connected together as close as possible to the DUT.

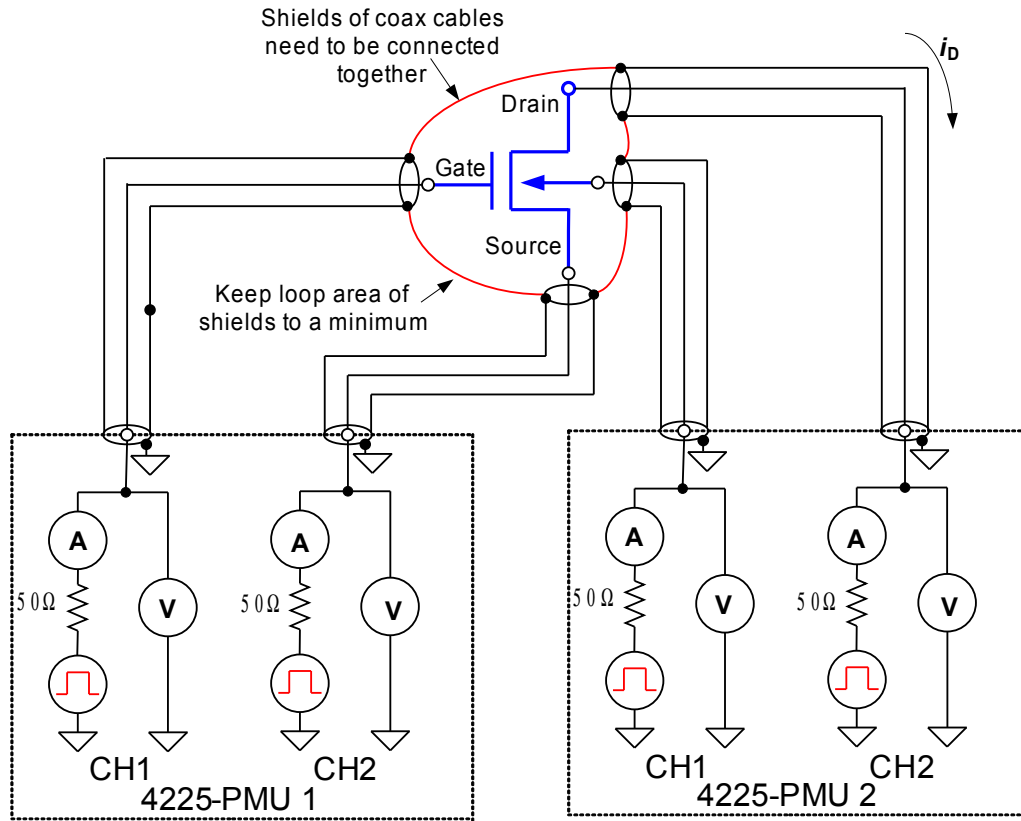


Figure 6. Connecting two PMUs to a four terminal MOSFET

MAKING CONNECTIONS TO THE MODEL 4225-PMU

Each channel of the 4225-PMU has a female SMA coax output connector. The center conductor of the SMA connector is the HI terminal of the module and outside shield is the AC common terminal. The Model 4225-PMU is supplied with cables (CA-404B) that are terminated with male SMA. The PMU also comes with cable assemblies (4200-PRB-C) that enable connections to SSMC connectors which are common for high frequency test signals. Table 1 lists the accessories that are supplied with the Model 4225-PMU.

Keithley Item #	Quantity	Description
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CA-404B	4	SMA to SMA 50 ohm cable, 2m
4200-PRB-C	2	SMA to SSMC Y-Cable Assembly, 6 inch

Table 1. Accessories supplied with the Model 4225-PMU Ultra Fast I-V Module

In some cases it may be necessary to connect the Model 4225-PMU to a prober or test fixture that may terminate in another type of connector such as BNC or triax. The Model 4200-PMU-PROBER-KIT is a collection of standard and custom connectors and accessories used to connect the Model 4225-PMU or Model 4220-PGU to a common variety of prober stations. Each kit contains all cables and accessories for one dual-channel Model 4225-PMU or Model 4220-PGU module. The contents of the Model 4200-PMU-PROBER_KIT are listed in Table 2 and are shown in the photograph of Figure 7.

Keithley Item #	Qty	Description	Notes
CA-19-2	2	BNC Male to BNC Male 50 Ω Cable, 1.5m	Useful for labs with significant BNC infrastructure
CAP-29	1	BNC Shorting Plug	Enables easier connection of outside shield of coax cable to manipulator probe
CA-452A	2	SMA to SMA Cable, 8 inch (20 cm)	Connection to SMU for sharing single SMA cable to device terminal
CS-565	3	BNC Female to BNC Female Adaptor	BNC barrels for BNC heavy labs
CS-712	3	BNC Female to Triax Male Adaptor (guard removed)	Adapt coax cable to BNC or to adapter to triax via CS-712
CS-1247	4	SMA Female to BNC Male Adaptor	Adapts SMA cable to BNC
CS-1252	2	SMA Male to BNC Female Adaptor	Adapts SMA outputs to BNC
CS-1390	2	Mini-Triax to Female SMA	Adapts SMU (no preamp) to SMA to allow sharing of a single SMA to DUT terminal
CS-1391	2	SMA Tee Adaptor, F-M-F	Allows for sharing of single

			SMA cable to DUT terminal.
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Table 2. Contents of Model 4200-PMU-PROBER-KIT

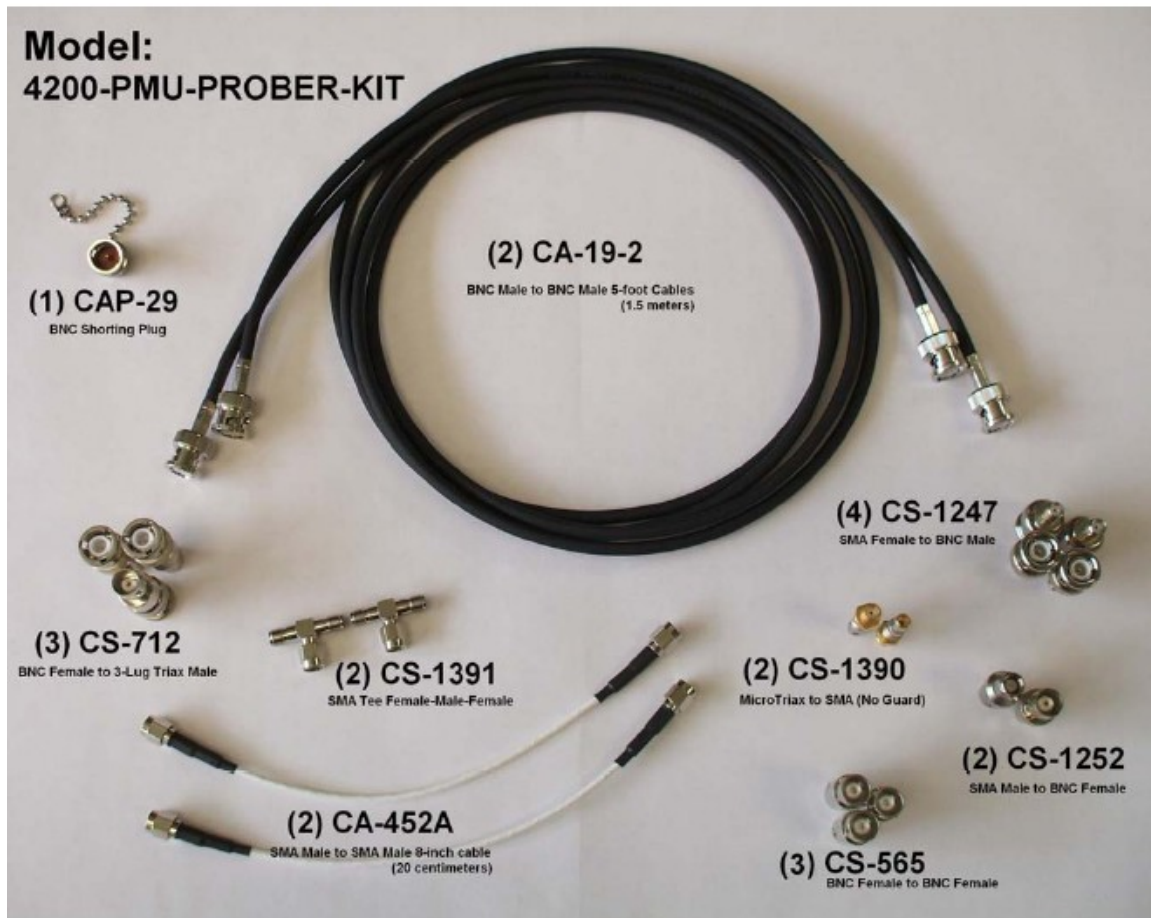


Figure 7. Contents of Model 4200-PMU-PROBER-KIT

The Model 4200-PMU-PROBER-KIT is useful for prober manipulators with triax or BNC connections. The following few sections provide detailed information on connecting the PMU to specific types of prober terminations including SSMC, SMA, BNC, triax, Cascade 12000 Series Prober, SUSS MicroTec PA200/300 Prober, and RF G-S-G probes.

PMU Connections to SSMC

SSMC is a type of coaxial probe holder connector that is rated from DC to about a gigahertz. SSMC connections are useful because they are small and threaded. A photograph of an American Probe and Technologies Series 74 SSMC probe holder is shown in Figure 8. These SSMC probe holders can be used on various manufacturers probe manipulators.

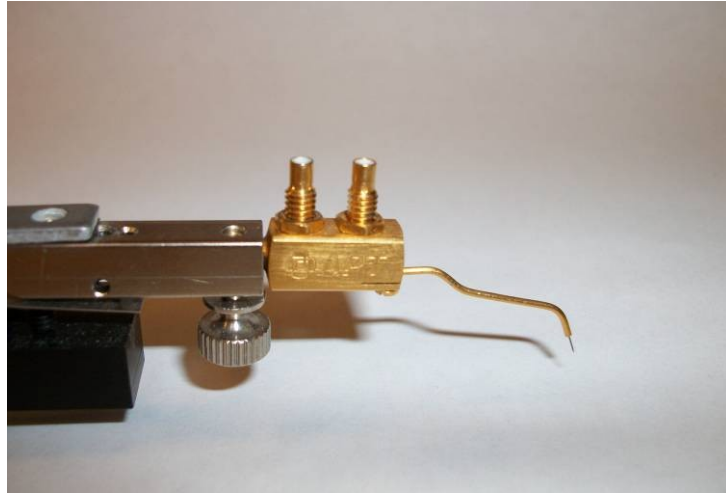


Figure 8. SSMC Probe Pin and Holder from American Probe and Technologies mounted on probe manipulator

Because the cable that comes with the PMU is terminated with SMA, an SMA to SSMC adaptor cable, Model 4200-PRB-C is included with the 4225-PMU to convert from SMA to dual SSMC connections on the DC manipulators. A photograph of this cable is shown in Figure 9. As shown, one of the SSMC male connectors provides the pulse signal from the center conductor of the PMU and the other SSMC male connector is connected to the shield of the coax cable and is the ground return path. From this ground return, a ground tap is provided to enable the user to easily connect the shields together as close as possible to the DUT.

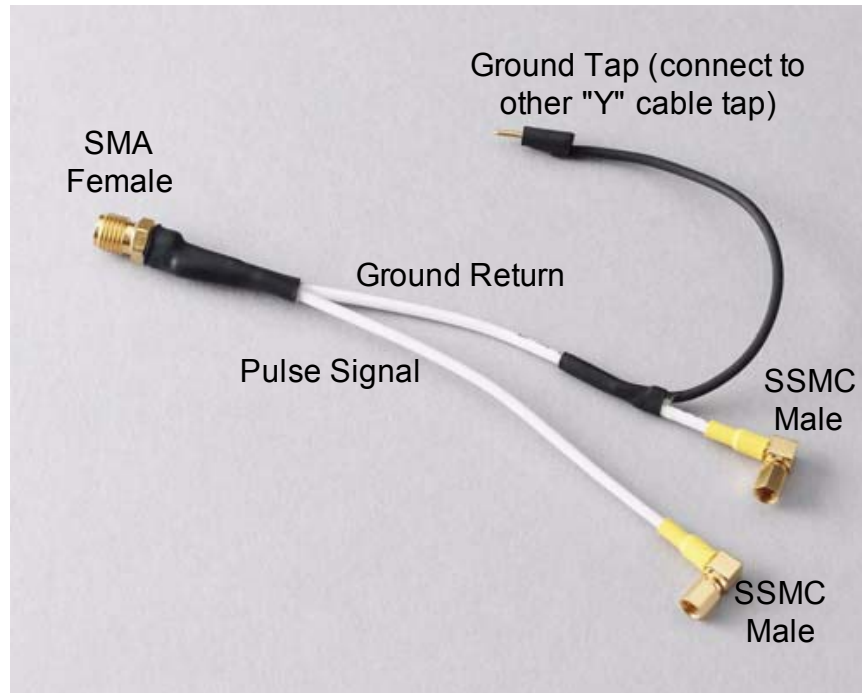


Figure 9. Model 4200-PRB-C SMA to SSMC Y-Cable Adaptor

Many DC probe manipulators are available with SSMC connections at the probe needle holder:

- Cascade DCM-2XX Kelvin DC probe manipulators
- Suss Microtec probe tips
- Signatone SCA-50 coaxial probes
- American Probe and Technologies 74 series coaxial probe holder

The 4200-PRB-C adaptor cables are appropriate for on-wafer pulse I-V testing of nominally DC structures. Figure 10 shows a pulse I-V connection from the PMU to the DC probes with SSMC connectors for a 4-terminal DC device structure using the 4200-PRB-C cables. Note the shields of the two channels are connected together using the ground taps of the 4200-PRB-C Y cables.

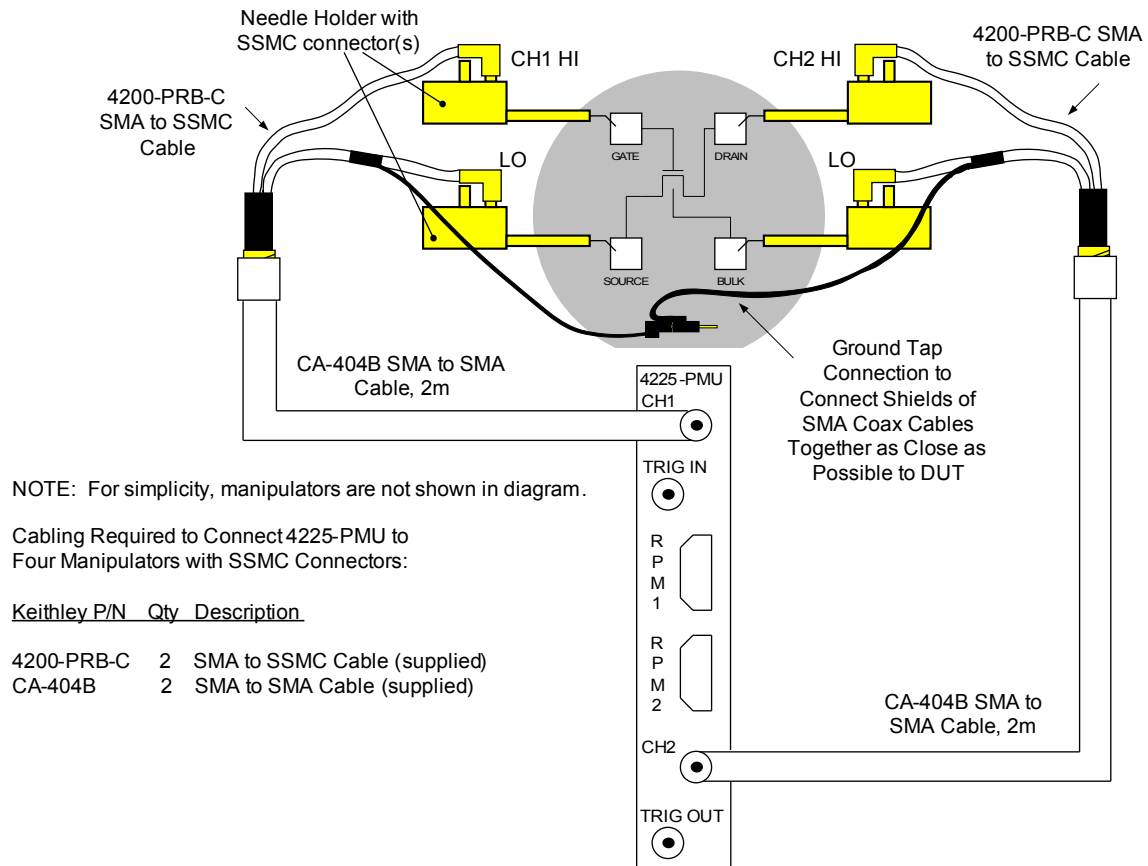


Figure 10. Cabling from 4225-PMU to DC manipulators with SSMC Connectors

Figure 11 shows the connections from the 4200-PRB-C Y cables to a 2-terminal device, a diode. In this case, the shield connections are made through the Ground Tap and the shield SSMC connectors of the cable are not connected to anything.

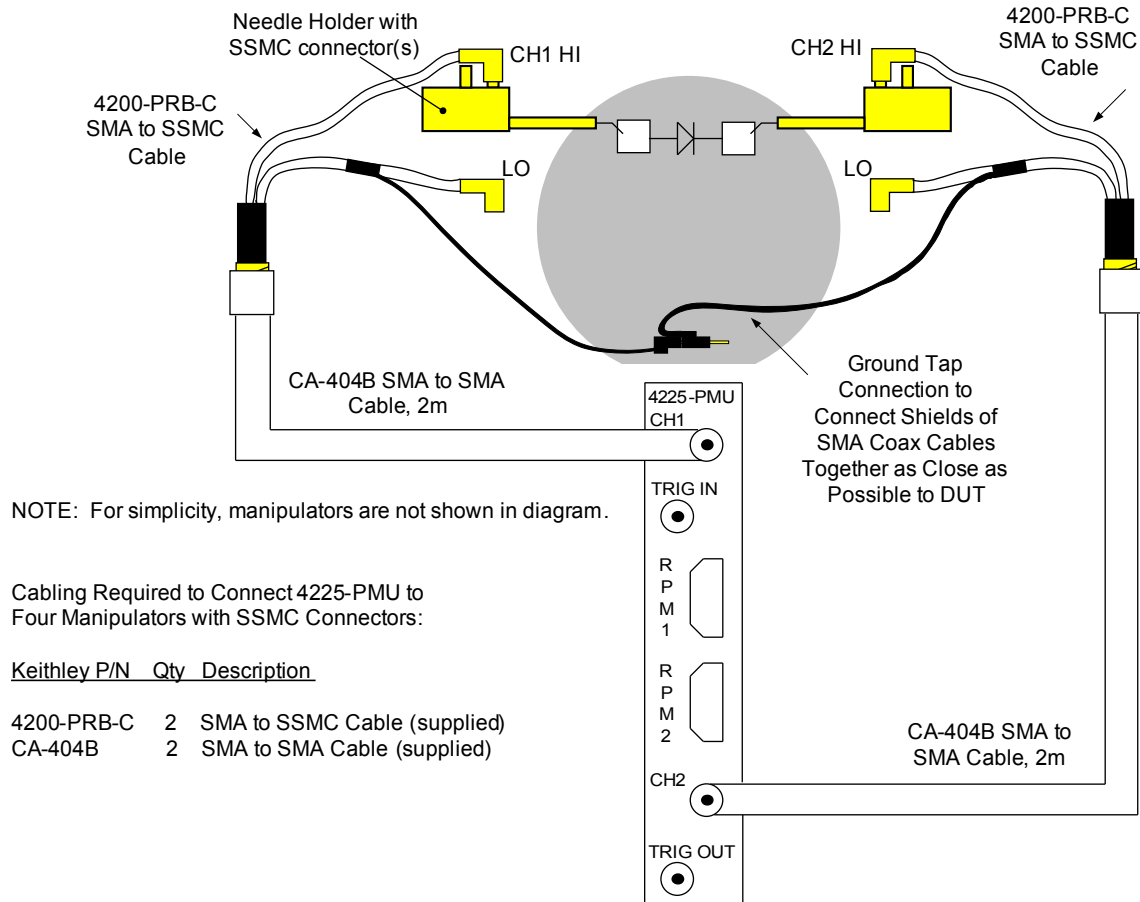


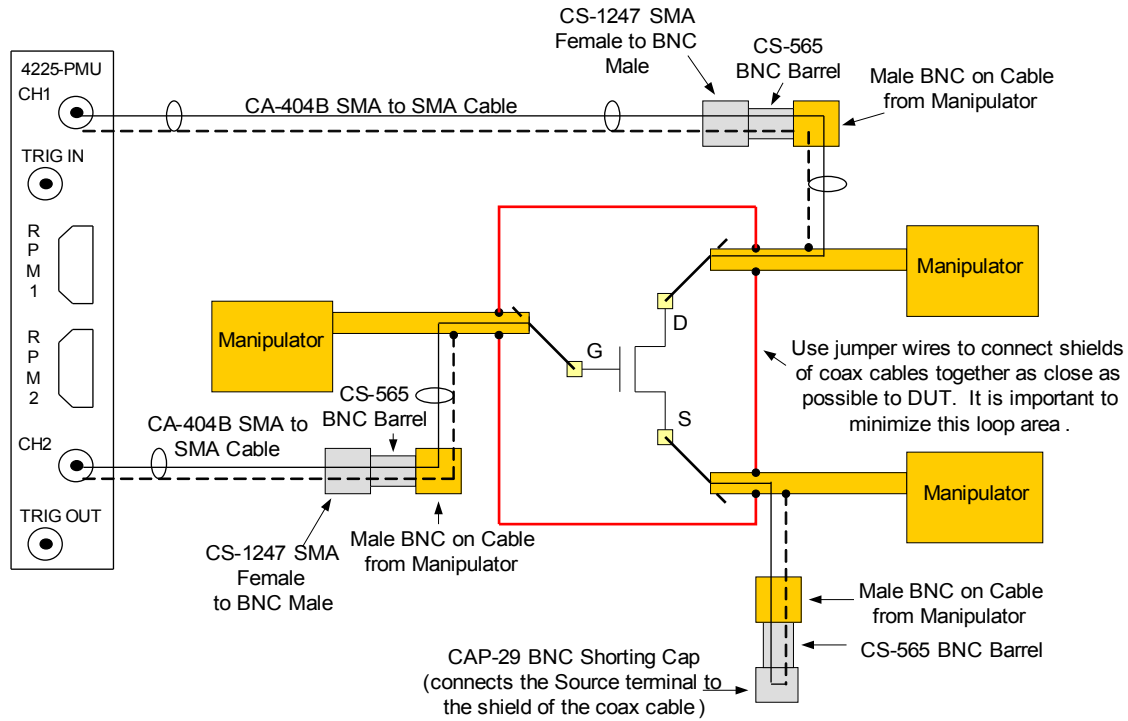
Figure 11. PMU Connections to a 2-terminal device using the 4200-PRB-C Y cables

PMU Connections to BNC

Probe manipulators can terminate with either male or female BNC. Figure 12 illustrates the connections from the 4225-PMU to a manipulator with male BNC connectors.

Adaptors are necessary to convert from the SMA cables that come with the PMU to the BNC connectors on the manipulators. The AC ground connection of the 4225-PMU is the outside shield of the coax cables. For the device Source terminal, the LO connection is made through the cabling to the manipulator using a BNC shorting cap.

To maintain signal integrity, it is very important to connect the shields of the manipulator probe holders together as close as possible to the device under test. This can be done with single insulated jumper wires with small alligator clips on either end. This connection will vary depending on the manufacturer of the manipulator.



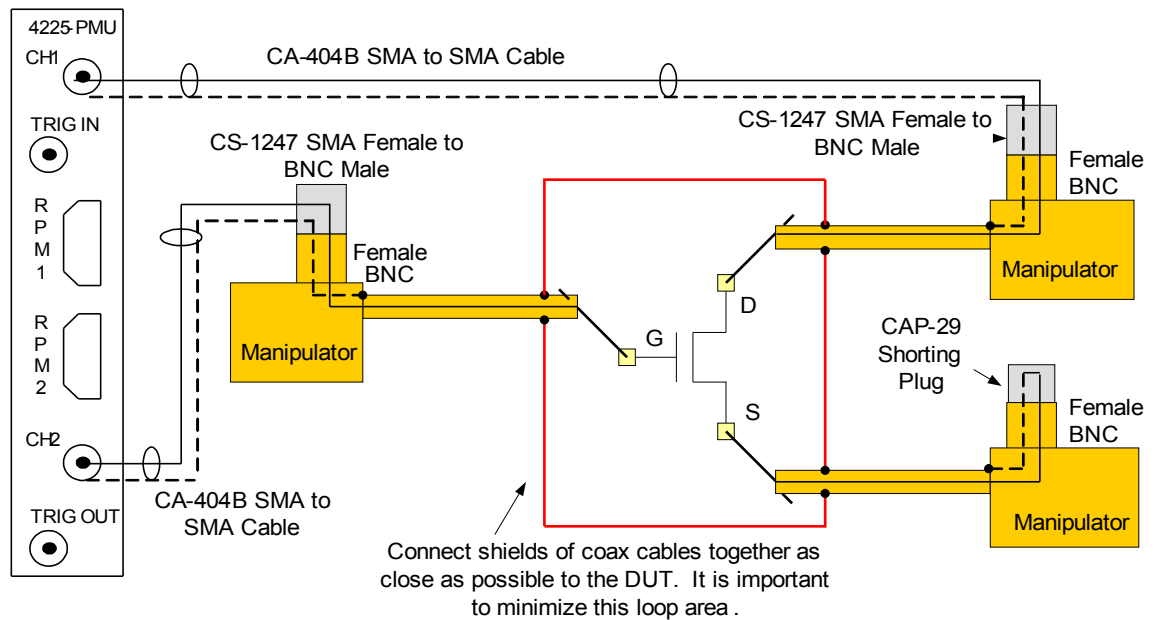
Cabling from 4225-PMU to 3 Manipulators terminated with Male BNC:

P/N	QTY	Description
CA-404B	2	SMA to SMA Cable, 2m (supplied)
CS-1247	2	SMA Female to BNC Male Adaptor*
CS-565	3	BNC Barrel*
CAP-29	1	BNC Shorting Plug*
N/A	3	Jumper Wires to Connect Shields (user supplied)

*included in Model 4200-PMU-PROBER-KIT

Figure 12. Connections from 4225-PMU to manipulators terminated with male BNC

Instead of male BNC connectors, some manipulators have female BNC connectors. For this case, Figure 13 shows the cabling from the 4225-PMU to manipulators with female BNC connectors along with a list of the required cables and adaptors to connect from SMA to BNC.



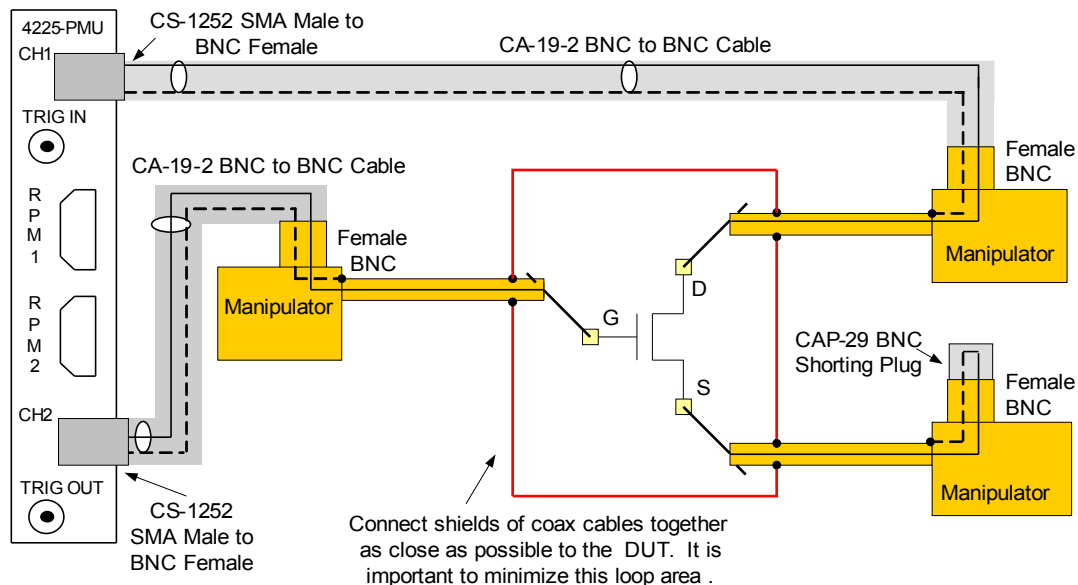
Cabling from 4225-PMU to 3 Manipulators Terminated with Female BNC:

Keithley P/N	QTY	Description
CA-404B	2	SMA to SMA Cable, 2m (supplied)
CS-1247	2	SMA Female to BNC Male Adaptor*
CAP-29	1	BNC Shorting Plug*
N/A	3	Jumper Wires to Connect Shields (user supplied)

*included in Model 4200-PMU-PROBER-KIT

Figure 13. PMU Connections to Prober Manipulators with Female BNC

Instead of using the SMA to SMA cables that are supplied with the 4225-PMU, it may be desirable to use BNC to BNC coax cables as shown in Figure 14. If this is the case, an SMA male to BNC female adaptor (CS-1252) is directly connected to the channel of the PMU and a BNC to BNC 50 ohm coax cable (CA-19-2) is connected to the BNC female connector on the manipulator or test fixture.



Cabling from 4225-PMU to 3 Manipulators Terminated with Female BNC:

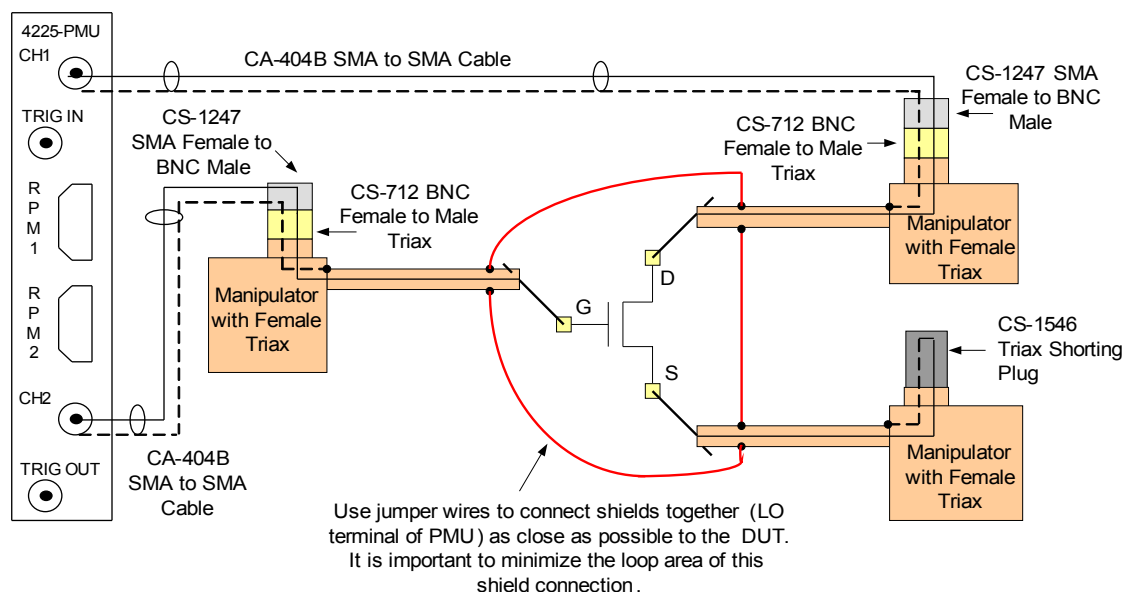
Keithley P/N	QTY	Description
CA-19-2	2	BNC to BNC cable, 50 ohm*
CS-1252	2	SMA Male to BNC Female Adaptor*
CAP-29	1	BNC Shorting Plug*
N/A	3	Jumper Wires to Connect Shields (user supplied)

*included in Model 4200-PMU-PROBER-KIT

Figure 14. PMU Connections to Prober using BNC Cables

PMU Connections to Female Triax

When connecting the SMA coax cables (2 conductors) from the PMU to a prober with triax connectors (3 conductors), it is best to use adaptors that eliminate the guard (inside shield) of the triax connector. Figure 15 shows this configuration using Keithley CS-712 Triax to BNC Adaptors which eliminate the inside shield (or guard) of the triax. Using these adaptors maintains the shield connections of the coax cables (LO terminal of the PMU) to the outside sleeves of each of the manipulators. The outside sleeves of each of the manipulators must be jumpered together as shown. The loop area of the jumpered shields must be kept to a minimum. The triax shorting plug connects the shield of the PMU to the probe connected to the Source terminal of the MOSFET.



Cabling from 4225-PMU to 3 Manipulators Terminated with Female Triax:

Keithley P/N	QTY	Description
CA-404B	2	SMA to SMA Cable, 2m (supplied)
CS-1247	2	SMA Female to BNC Male Adaptor*
CS-712	2	BNC Female to Triax Male Adaptor (guard removed)*
CS-1546	1	Triax Shorting Plug
N/A	3	Jumper Wires to Connect Shields (user supplied)

*included in Model 4200-PMU-PROBER-KIT

Figure 15. Cabling from Model 4225-PMU to Prober Manipulators with Female Triax

PMU Connections to the Cascade 12000 Series Prober

The Keithley Model 4210-MMPC-C Multi-Measurement Prober Cable Kit is a collection of standard and custom connectors and accessories that is used to make DC I-V, C-V, and pulsed I-V measurements using a single prober cable set-up. This kit has been assembled with all the cables and adaptors necessary to connect any of the 4200 modules including the Model 4225-PMU to the Cascade Microtech 12000 Prober Series (manipulator type DCM-200 series with dual SSMC probe tips). Figure 16 shows the accessories that are included in the Model 4210-MMPC-C Cable Kit. The blue MMPC cables that come in the kit are special triax cables that allow both high precision DC and high bandwidth AC connections to the probes and DUT. More importantly, the user doesn't need to disturb the cabling to the probes; you need only connect the desired instrument to the connector

bulkhead of the prober.

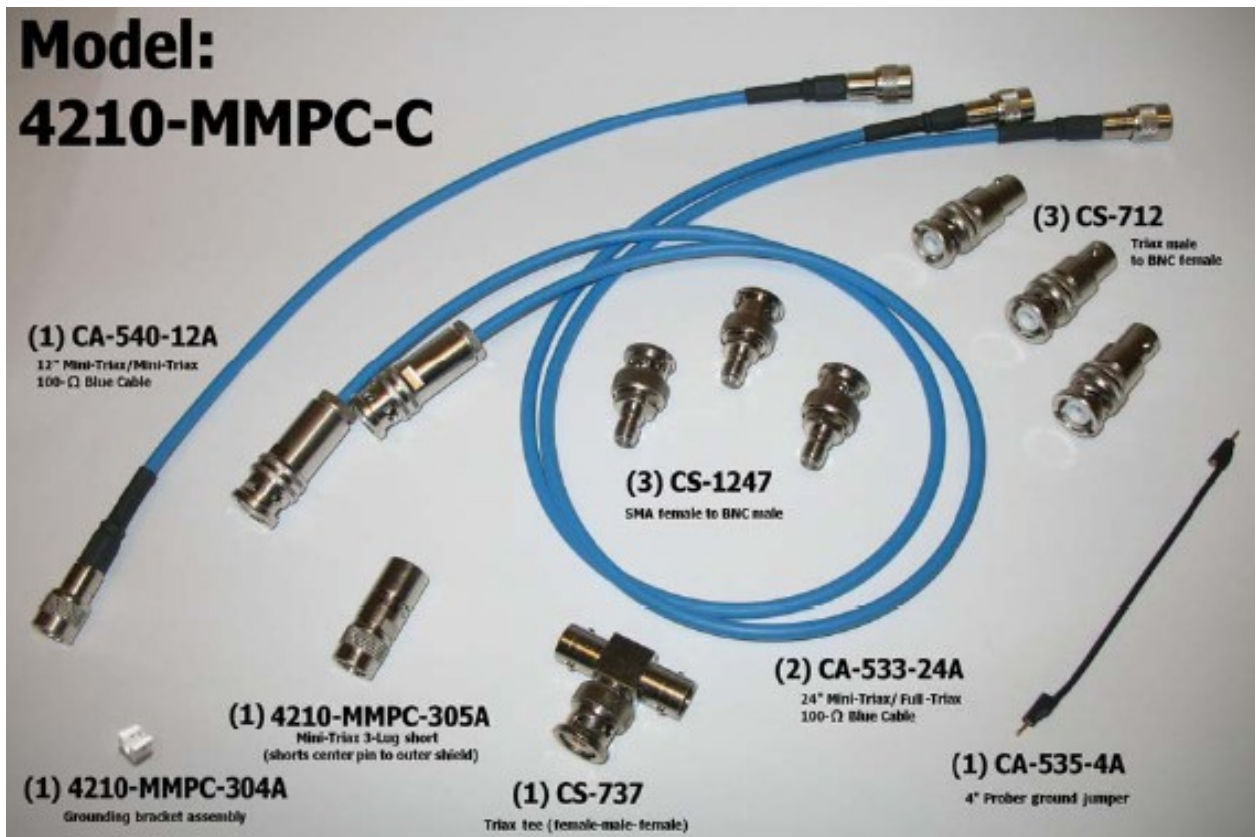


Figure 16. Model 4210-MMPC-C Multi-Measurement Prober Cable Kit

A diagram showing the connections from the Model 4225-PMU to the Cascade prober connected to a 3-terminal MOSfet is shown in Figure 17. The connections shown in this diagram requires two of the Model 4210-MMPC-C Cable Kits. The Cascade manipulators have female mini-triax connectors. The blue CA-533-24A cables connect

from the full size female triax bulkhead on the patch panel to the mini-triax on the manipulator.

The 4210-MMPC-305 Mini-Triax Shorting Plug connects the center conductor of the triax cable to the outer shield. This connects the Source terminal of the MOSFET to the PMU LO terminal. The 4210-MMPC-304A Grounding Bracket Assembly enables the user to easily connect the shields of each cable together using the CA-535-4A Ground Jumpers as shown.

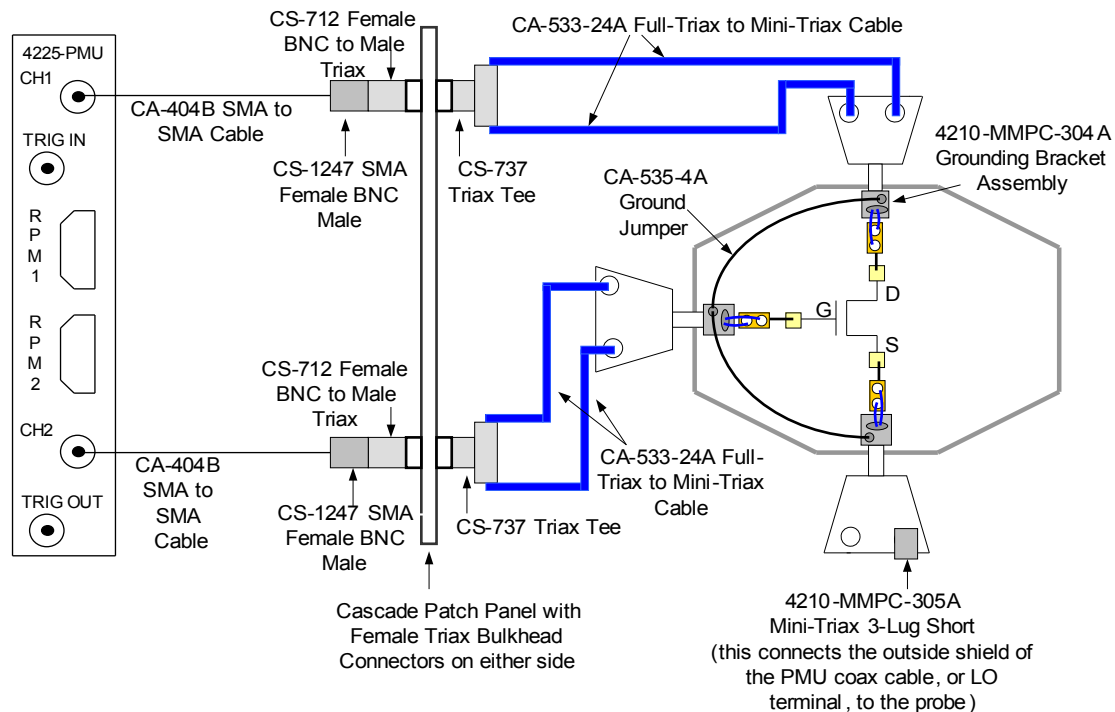


Figure 17. 4225-PMU Connections to the Cascade Prober
Using the 4210-MMPC-C Multi-Measurement Prober Cable Kit

PMU Connections to the SUSS MicroTec PA200/300 Prober Series

The Model 4210-MMPC-S Multi-Measurement Prober Cable Kit is a collection of standard and custom connectors and accessories that can be used for DC I-V, C-V, and pulsed I-V measurements using the SUSS MicroTec PA200/300 prober series. This high performance triaxial cable kit enables the user to easily switch between different test signals, DC I-V, C-V, and pulsed I-V, without having to change the probe manipulator cabling.

Figure 18 shows the contents of the 4210-MMPC-S Cable Kit. Notice that each kit contains a MMPC Prober Cable Assembly, CA-532, which easily connects to the SUSS manipulators. Besides enabling easy connections between the PMU and the probe tips, this cable assembly enables the user to connect the shields together close to the DUT. Each kit is designed to contain all cables and accessories for pulsing and measuring to one pin. So, a two-pin test set-up usually requires two Model 4210-MMPC-S kits.



Figure 18. Contents of Model 4210-MMPC-S Multi-Measurement Prober Cable Kit

Figure 19 shows the PMU connected to a 3-terminal MOSFET using three of the 4210-MMPC-S cable kits. In this drawing, CH1 and CH2 of the PMU are connected to

the Drain and Gate terminals of the MOSFET using the 4210-MMPC-S cable kits. The Source terminal of the MOSFET is connected to the LO terminal (outside shield of the coax cable) using the CS-1546 Triax Shorting Cap that comes with the 4210-MMPC-S kit. One kit is required for each pin on the device.

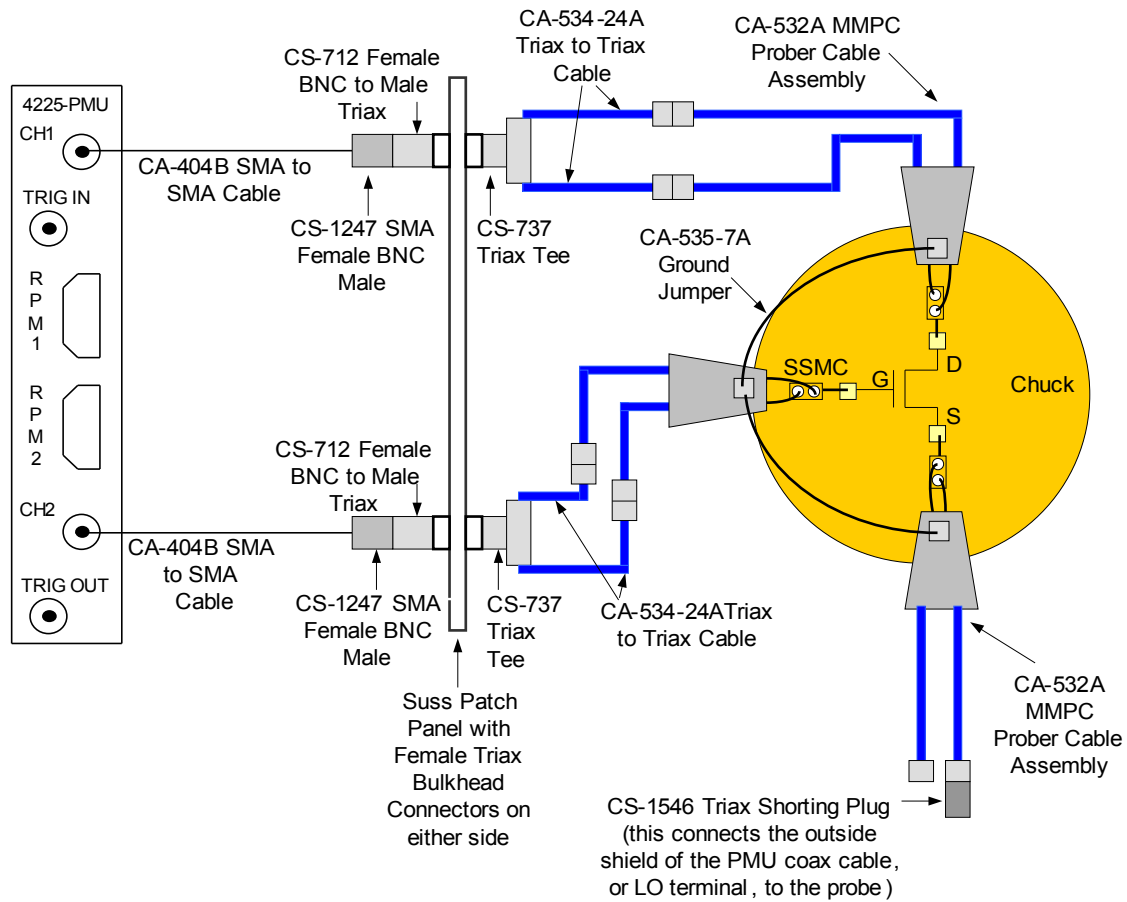


Figure 19. Schematic of PMU Connections Using Three Model 4210-MMPC-S Cable Kits

PMU Connections to an RF prober

If the device has an RF layout (G-S-G), the supplied Y cables and DC probe manipulators will most likely not be sufficient since most RF G-S-G probes have SMA connectors. If this is the case, use the supplied CA-404B SMA cables to connect from the PMU output to the RF manipulators as shown in Figure 20, using an appropriate adaptor or “connector

Diagram illustrating the cabling required to connect a Keithley 4225-PMU to two RF G-S-G Probes.

The setup includes two RF G-S-G Probes connected to a central device (likely a transistor) with GATE, DRAIN, and SOURCE terminals. The probes are connected via CA-404B SMA to SMA Cables (2m, supplied with PMU) and Connector Savers.

The cables are connected to the 4225-PMU's CH1 and CH2 ports.

Cabling Required to Connect 4225-PMU to Two RF G-S-G Probes:

Keithley P/N	Qty	Description
CA-404B	2	SMA to SMA Cable (supplied)
N/A	2	Connector Savers (user supplied)

Note: The diagram also shows TRIG IN and TRIG OUT ports on the 4225-PMU, which are connected to the probes.

A photograph showing the SMA cables from the PMU to an RF prober with connector savers is shown in Figure 21.

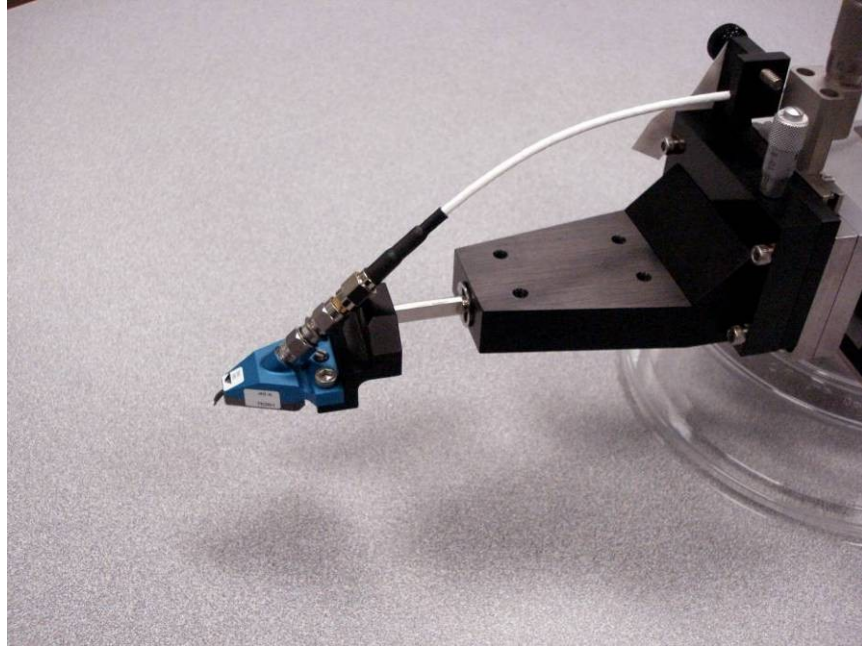


Figure 21. SMA Connection to RF Prober (with connector saver)

Summary of Cabling Requirements to Model 4225-PMU

Table 3 summarizes the various DUT terminations and the corresponding cabling to the 4225-PMU. This table does not include the connections to the shields of the coax cabling as previously described in the text. All part numbers listed in the table are from Keithley Instruments.

Test Fixture or Prober Terminations	Required Cabling for Connections to Model 4225-PMU (female SMA)
SSMC	SMA Cables and 4200-PRB-C Cables (supplied with PMU)
SMA	SMA Cables (supplied with PMU)
BNC	4200-PMU-PROBER-KIT Cable Kit
Triax	4200-PMU-PROBER-KIT Cable Kit
Cascade 12000 Series Prober	4210-MMPC-C Cable Kit
SUSS MicroTec PA200/300 Prober	4210-MMPC-S Cable Kit
RF G-S-G	SMA Cables (supplied) with a user supplied connector saver

Table 3. Summary of manipulator connector types and corresponding cabling and adapters.

CABLING TO THE MODEL 4225-RPM REMOTE AMPLIFIER/SWITCH

The 4225-RPM Remote Amplifier/Switch has two functions, one to extend the low current measurement ranges of the PMU and the other to enable switching between the 4225-PMU, 4210-CVU, and the 4200-SMU. The 4225-RPM is connected to the PMU using the supplied RPM Interconnect Cable. Figure 22 is a photograph of two 4225-RPM modules connected to the two channels of the Model 4225-PMU. The RPM comes with a magnetic platen mounting base which allows the RPM to be located next to the manipulators on the chuck platen. This minimizes measurement problems caused by long cable lengths.

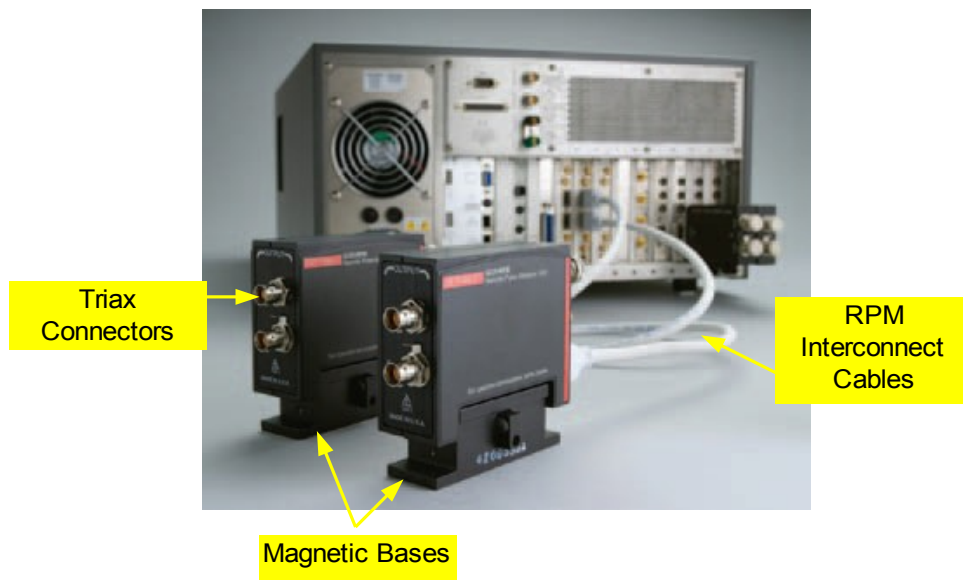


Figure 22. 4225-RPM connections to the 4225-PMU in the 4200-SCS Chassis

Unlike the SMA output connectors of the Model 4225-PMU, the Model 4225-RPM has two female triax connectors as outputs, one for Force and one for Sense. The Input and Output connectors of the 4225-RPM are shown in Figure 23. Notice the HI terminal of the PMU is on the center conductor of the triax output terminal of the RPM and the LO terminals of the PMU is on the outside shield of the triax output connector of the RPM. The Input connections are as follows: two female triax connectors for the Force and Sense terminals of the SMU, two female SMA connectors for the CURR and POT terminals of the CVU, and one connector for the RPM Interconnect Cable.

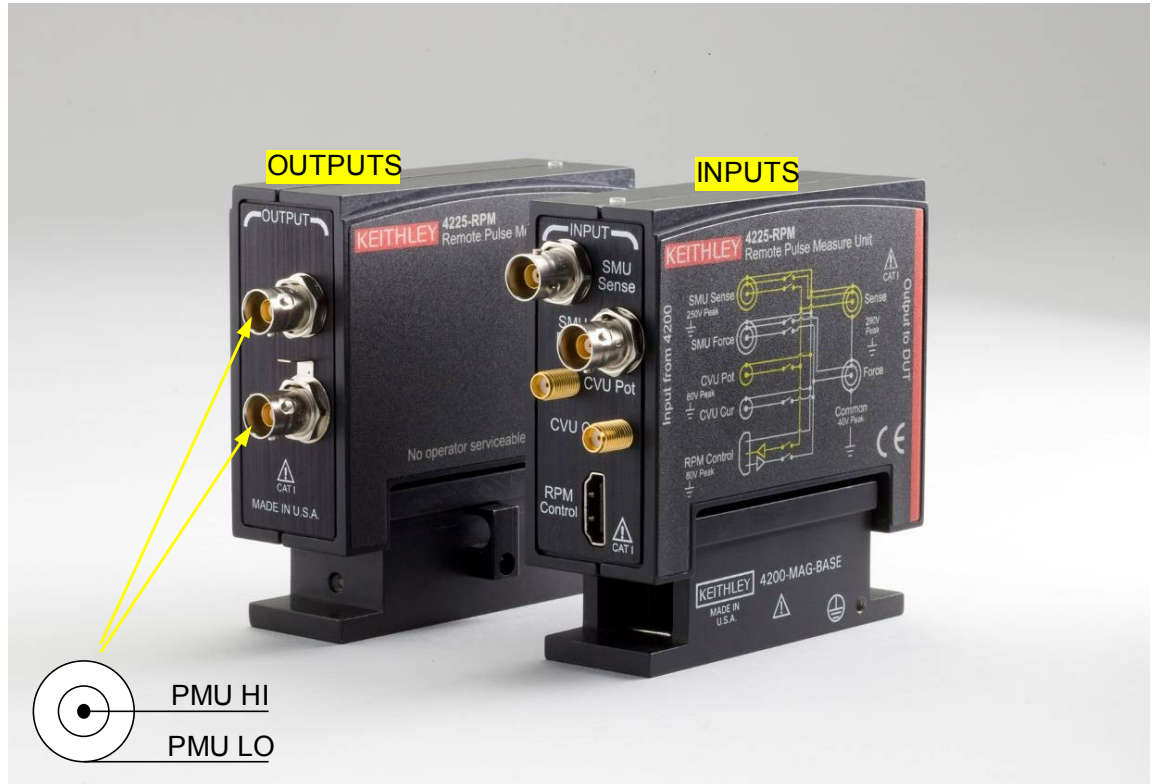


Figure 23. Input and Output connections of the 4225-RPM

Using the cables and adaptors that are supplied with both the 4225-PMU and the 4225-RPM, the user can make connections to either SMA or SSMC. Table 4 lists the accessories supplied with the 4225-RPM and a photograph of these accessories is shown in Figure 24. In addition to cables and adaptors, the RPM is supplied with Model 4200-MAG-BASE Magnetic Preamp Base which allows the RPM to be located next to the manipulators on the chuck platen.

Keithley Item #	Quantity	Description
CA-547-2A	1	RPM Interconnect Cable, 6 feet
CS-712	1	Male Triax to Female BNC Adaptor, guard removed
CS-1247	1	BNC Male to SMA Female Adaptor
CA-452A	1	SMA to SMA Cable, 8 inch
4200-MAG-BASE	1	Magnetic Preamp Base
CAP-31	4	Triax Caps (for shielding, not shorting)
CAP-47	2	SMA Male 50 ohm cap

Table 4. Accessories supplied with the Model 4225-RPM

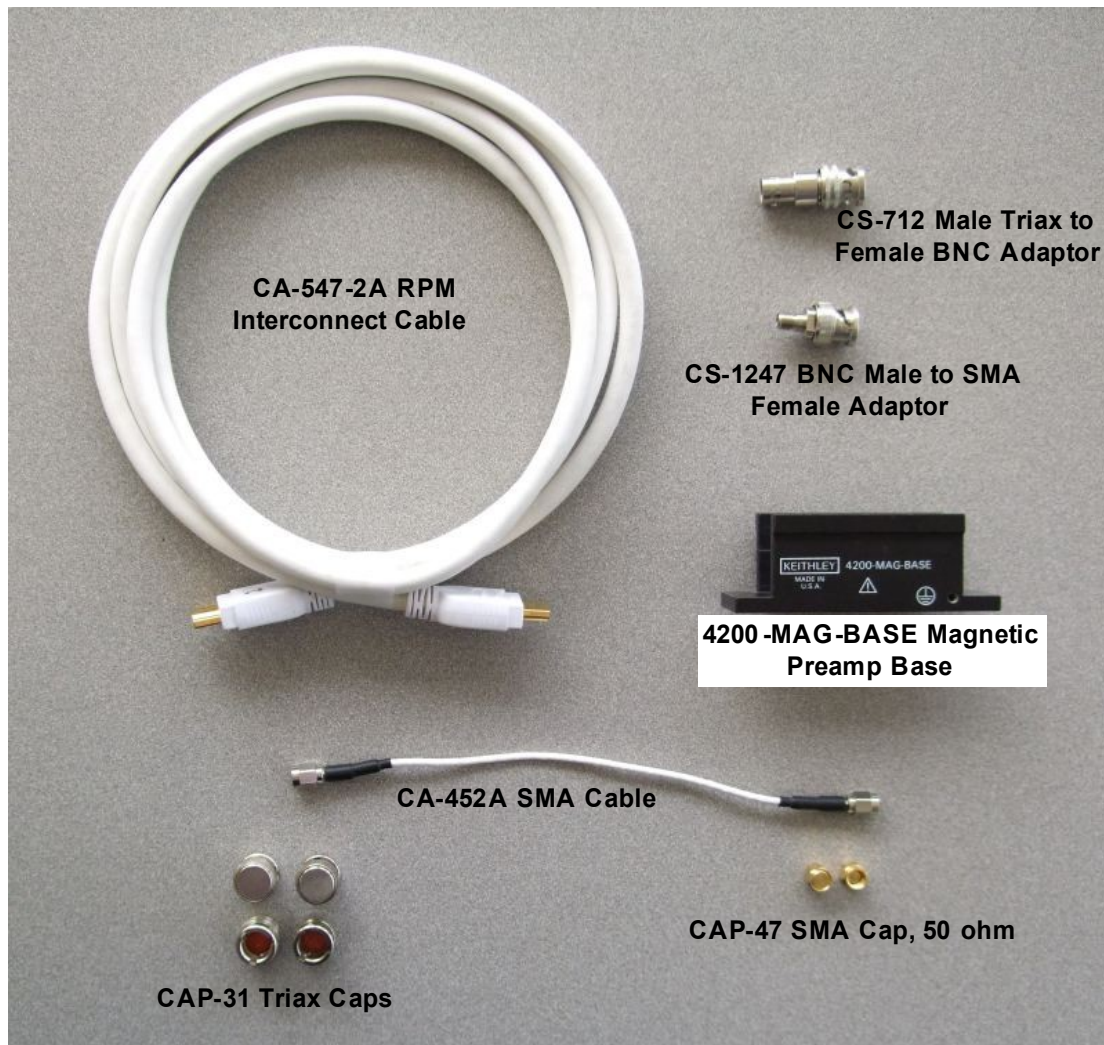


Figure 24. Photograph of accessories supplied with Model 4225-RPM

The photograph in Figure 25 shows the connections from the female triax connectors on the RPM to male SMA using the supplied accessories. It is important to keep the cable length from the output of the RPM as short as possible. The RPM is supplied with Model 4200-MAG-BASE Magnetic Preamp Base which allows the RPM to be located next to the manipulators on the chuck platen. The following few paragraphs will explain how to make connections from the output of the RPM to various prober terminations.

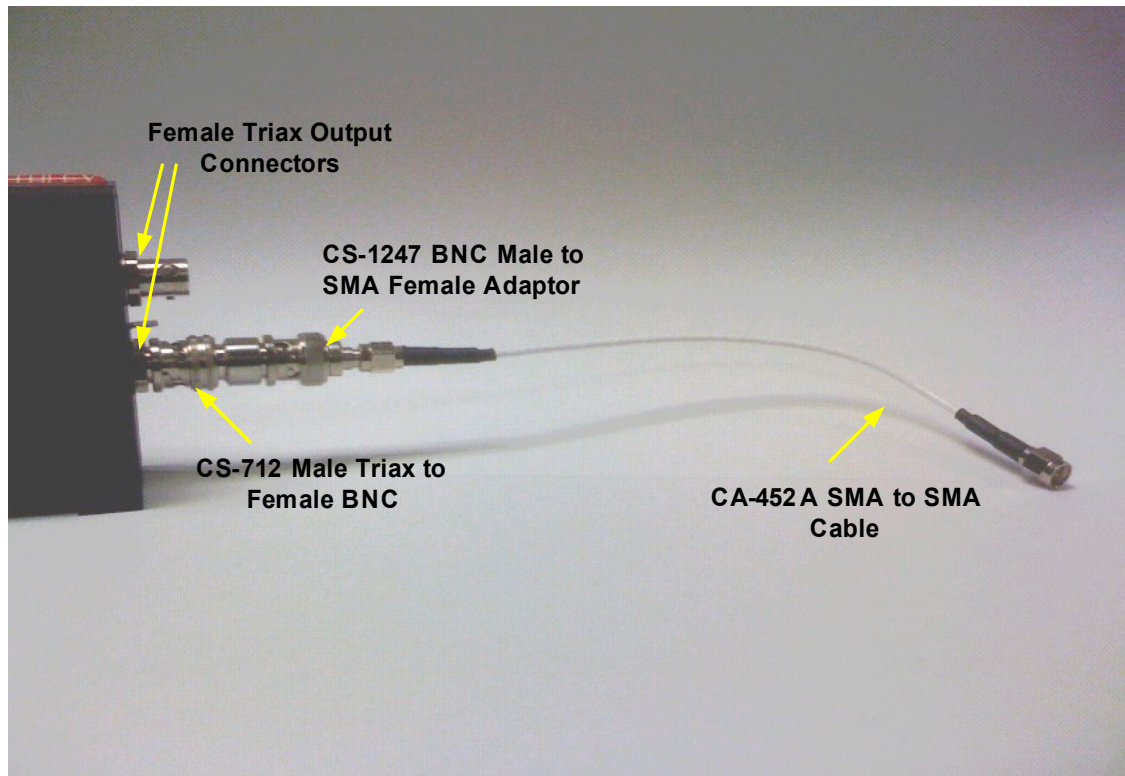


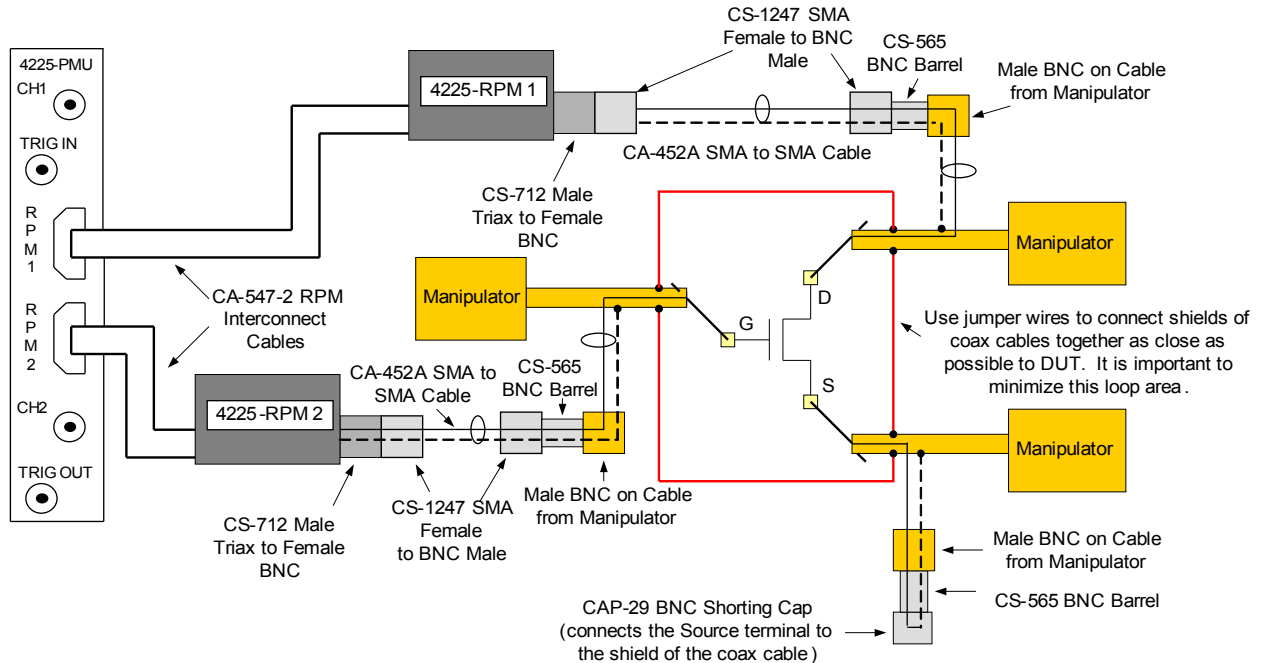
Figure 25. Connections to RPM using supplied cabling

RPM Connections to SMA or SSMC

As mentioned previously, the PMU and RPM include the adaptors and cabling to connect to either SMA or SSMC. For SSMC just add the 4200-PRB-C SMA to SSMC Y Cable and make sure the shields are connected together using the Ground Taps.

RPM Connections to BNC

Figure 26 illustrates the connections from the output of the 4225-RPM to a manipulator with male BNC connectors. In place of the SMA Cables and SMA to BNC adaptors, a short (1 or 2 foot) 50 ohm BNC to BNC cable can be used to simplify the connections to the manipulator. Keeping the cable from the output of the RPM to the manipulator as short as possible is important in order to achieve the highest frequency signal output.



Cabling from 4225-PMU with Two 4225-RPMs to 3 Manipulators Terminated with Male BNC:

P/N	QTY	Description
CA-547-2	2	RPM Interconnect Cable, 6 feet (one supplied with each RPM)
CA-452A	2	SMA to SMA Cable, 8 inch (one supplied with each RPM)
CS-1247	4	SMA Female to BNC Male Adaptor (one supplied with each RPM)*
CS-712	2	Triax Male to BNC Female Adaptor, guard removed (one supplied with each RPM)
CS-565	3	BNC Barrel*
CAP-29	1	BNC Shorting Plug*
N/A	3	Jumper Wires to Connect Shields (user supplied)

*included in Model 4200-PMU-PROBER-KIT

Figure 26. Connections from 4225-RPMs to manipulators terminated with male BNC

RPM Connections to Triax

From the output of the RPM to a test fixture or manipulator with female triax connectors, the cabling depends on the application. Here are three possibilities:

1. Use the cables and adaptors that come with the RPM that enable termination to BNC. Then use an additional CS-1247 SMA Female to BNC male adaptor and an additional CS-712 Triax Male to BNC Female Adaptor. This connection is good for higher frequencies, but will not be optimal for higher impedance or lower current measurements with the SMU through the RPM.
2. Use Model 7078-TRX-1 Triax to Triax Cable, 1 foot. This cable is good for high impedance measurements, but not high frequency measurements: pulse widths

- <1 μ s or higher frequency. This is not optimal for high frequency CV measurements through the RPM.
3. Use two CA-534-24A Blue Triax Cables and one CS-737 Triax Tee. This combination is good for high impedance and high frequency, low current, and CV measurements provided the distance from the tee to the probe tip is short (<6 to 10 inches). This is because of the need to tie the grounds together.

RPM Connections to the Cascade 12000 Series Prober

The Model 4210-MMPC-C Multi-Measurement Prober Cable Kit comes with all the accessories required to connect one PMU channel to one Cascade Manipulator. The contents of this kit were shown in Figure 16 (this was previously shown). Figure 27 shows multiple 4210-MMPC-C cable kit connections to the Cascade Manipulators. Notice the blue triax cables connect directly from the RPM to the mini-triax connectors on the probe manipulators, bypassing the patch panel. Ground jumpers and a grounding bracket assembly are supplied in the kit to ensure the shields are connected together as close as possible to the DUT. Figure 28 shows the inside of the EMI shield (top hat), viewed from the top where the microscope is normally positioned.



Figure 27. Multiple RPMs connected to Cascade manipulators

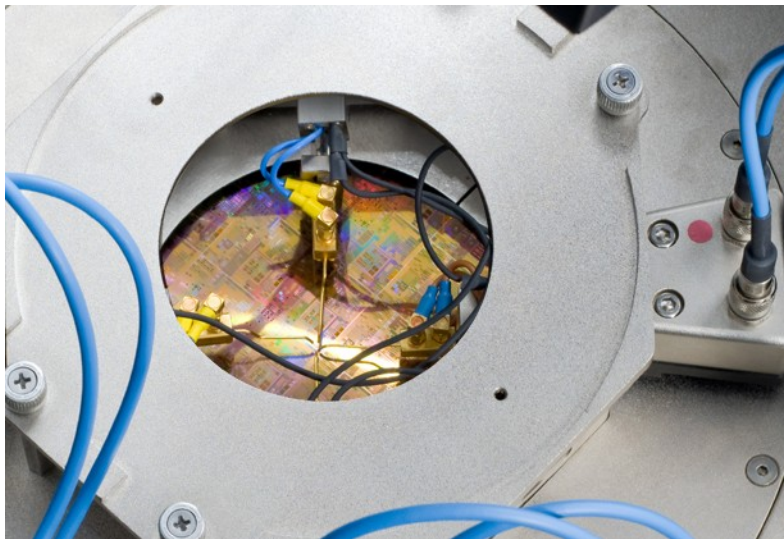


Figure 28. View inside of the Cascade microscope access port

RPM Connections to the SUSS MicroTec PA200/300 Prober Series

The Model 4210-MMPC-S Multi-Measurement Prober Cable Kit comes with all the accessories required to connect one PMU channel to one SUSS manipulator. Figure 29

shows the connections of the PMU to a three-terminal MOSFET device. The RPM Interconnect cables are fed through the feedthru of the bulkhead shield of the SUSS prober. The magnetic bases that come with the RPMs enable the RPMs to rest inside the shield of the prober so they can be located close to the DUT.

Notice CH1 of the PMU is connected to the Drain, CH2 is connected to the Gate, and the shield (LO terminal) is connected to the Source. The LO terminal connection is made using a triax shorting cap which shorts the shield to the probe tip.

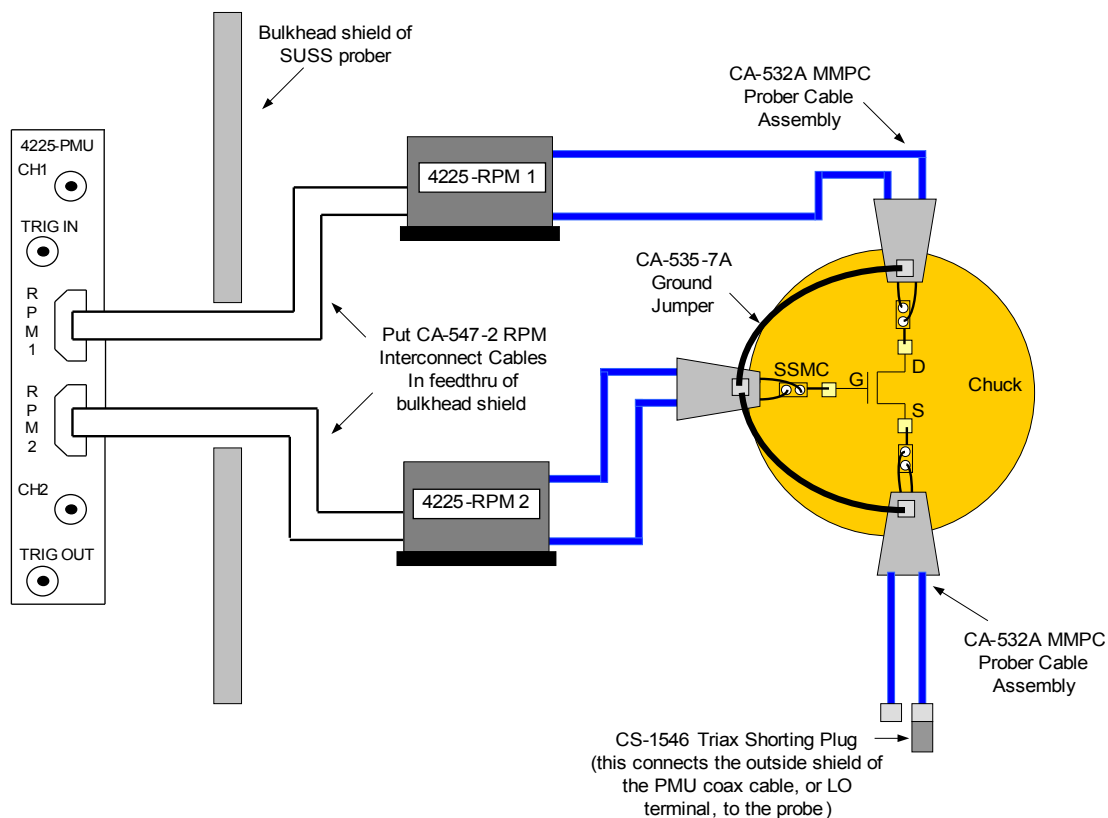


Figure 29. Connections of the 4225-RPM to the SUSS Prober using three Model 4210-MMPC-S Cable Kits

Table 5 provides a summary of RPM connections based on terminations to the DUT:

Test Fixture or	Required Cabling for Connections from Output of One Model
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Prober Terminations	4225-RPM (female triax)
SSMC	CS-712 Male Triax to Female BNC Adaptor with guard removed (supplied), SC-1247 Male BNC to Female SMA (supplied), CA-452A SMA to SMA Cable, 8 in. (supplied), and 4200-PRB-C SMA to SSMC Cable (supplied with PMU)
BNC	CS-712 Male Triax to Female BNC Adaptor with guard removed (supplied) and CA-19-2 BNC to BNC Coax Cable, 5ft. (50 ohm)
SMA	CS-712 Male Triax to Female BNC Adaptor with guard removed (supplied), SC-1247 Male BNC to Female SMA, CA-452A SMA to SMA Cable, 8 in. (supplied)
Triax	1. Use cables and adaptors supplied with RPM with an additional CS-712 Triax Male to BNC Female Adaptor and an additional CS-1247 SMA Female to BNC Male Adaptor 2. Use 7078-TRX-1 Triax to Triax Cable, 1 ft. 3. Use two CA-534-24A Triax to Triax Cables and one CS-737 Triax Tee
Triax – Cascade MircoTech 12000 Series	Model 4210-MMPC-C Multi-Measurement Prober Cable Kit
Triax – Suss MicroTech PA300 Prober Series	Model 4210-MMPC-S Multi-Measurement Prober Cable Kit

Table 5. RPM Connections

RPM Connections When Using the 4225-RPM as a Switch

The Model 4225-RPM can be used to switch the SMUs, CVU, and PMU to the device under test. Each instrument includes the necessary cabling to connect from its output to the input of the RPM. Figure 30 shows an example of the outputs of two 4200-SMUS, the 4210-CVU, and the 4225-PMU connected to the inputs of two 4225-RPMs. In this example, a four wire measurement is made from the outputs of the SMUs and the CVU to the RPM inputs. However, in this case, the cabling at the output of the RPM is

configured for two-wire. With the appropriate cabling, a four-wire measurement can be made to the device under test since the RPM has both Force and Sense output terminals.

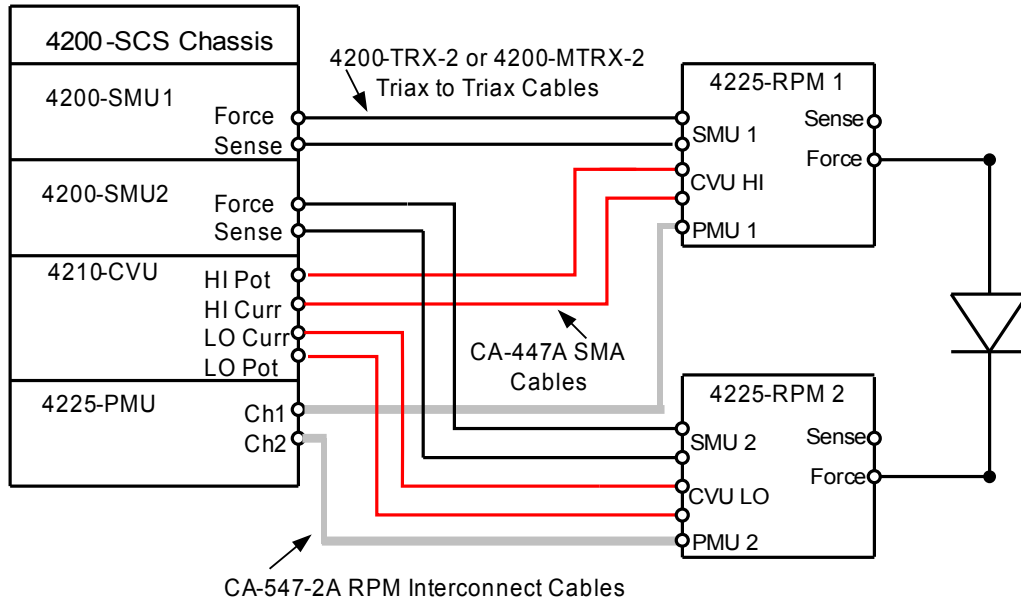


Figure 30. Using the 4225-RPM as a Switch

Another example of using the 4225-RPM as a switch is shown in Figure 31. In this case the SMUs, CVU, and PMU are switched through the RPM to a Suss prober using the Model 4210-MMPC-S Cable Kits.

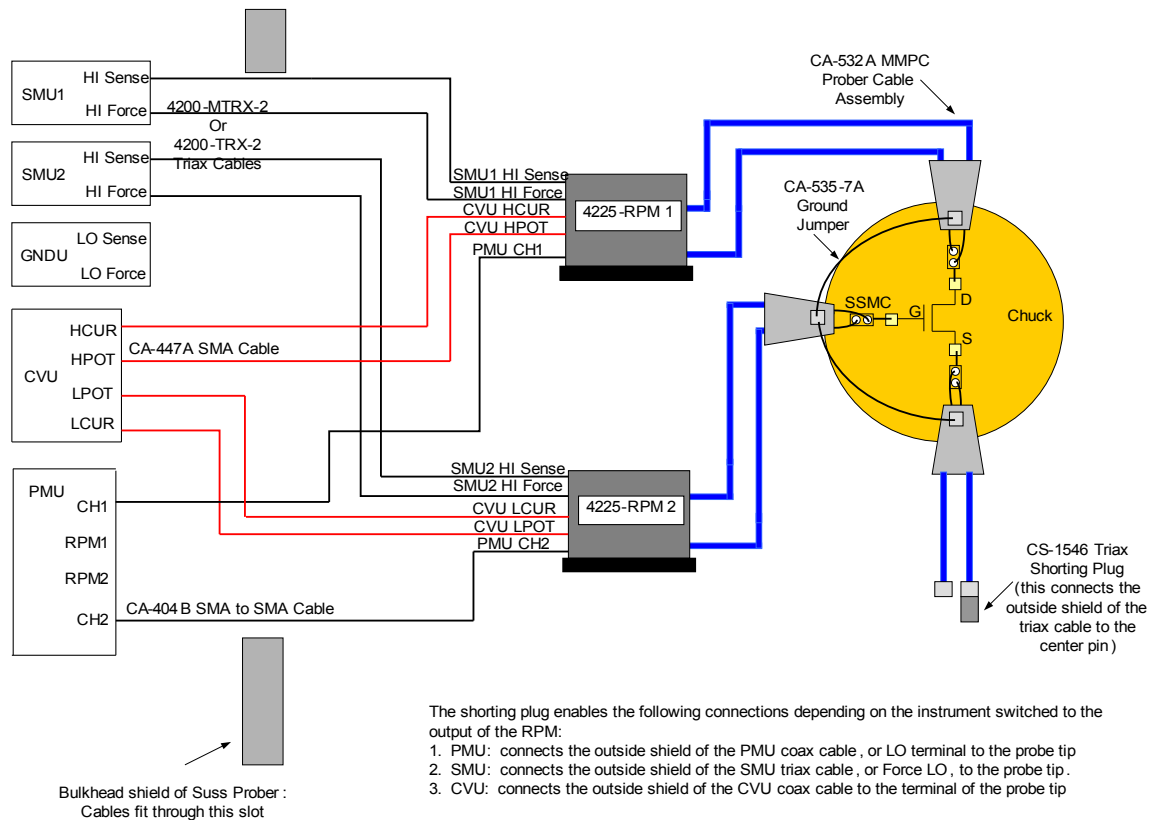


Figure 31. Using the 4225-RPMs as a Switch to the Suss Prober

OPTIMIZING CONNECTIONS FOR ULTRA FAST I-V

Making proper connections to the device under test is essential to making optimal ultra fast I-V measurements. Some of these include choosing the correct cables and appropriately connecting the shields. The previous sections described specific cabling configurations for various probe terminations. Since not all possible cabling configurations can be illustrated as examples, here are some cabling and connections rules to optimize ultra fast I-V testing:

Use cables, connectors, and probers rated for high frequency:

For pulsing and measuring at about 1 μ s pulse widths and longer, standard 50 ohm coax cables and DC probe manipulators should be sufficient. For test signals above 1 MHz (<1 μ s pulse widths), high frequency cables and connections become important. Ensure the DC probe manipulators are rated to at least 150 MHz.

Properly connect the shields of the coax cables:

Ideally the low side of the device under test should be tied to the shield of the coax cable. Do not connect the low side of the device to the DC ground (4200-GNDU common) because it will limit the fidelity of the pulse resulting in a distorted pulse and/or signal losses. Remember, AC signals need AC return paths, and DC signals need DC return paths. Even though connecting the low side of the DUT to the 4200-GNDU will work, the return signal now may follow two paths back to the instrument. One path will be through earth ground and the second will be through capacitive couple to the coax shield.

For multiple PMU channels, the shields from all PMU channels in the test should be connected together as close as possible to the DUT.

Minimize Loop Area of Shield Connections

For pulses with rise times less than a microsecond, it becomes important to minimize the inductance due to the cabling in the test circuit. Fast rise times and large inductances can cause large voltage overshoots, oscillations, and ringing in high speed pulsed measurement circuits. This phenomenon is based on Lenz's Law: $V = L \, di/dt$.

To reduce the effects of inductance, minimize the loop area of the shield connections. From the point the shield breaks away from the cable, keep the cable length to a minimum. This loop area is illustrated in Figure 32.

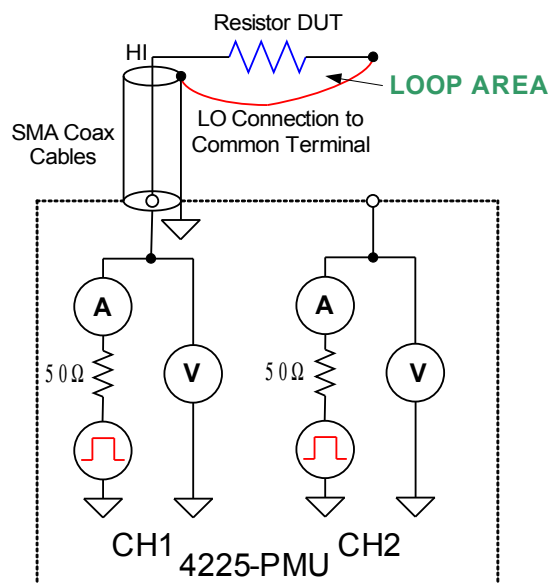


Figure 32. Minimize Loop Area between the HI and LO Terminal Connections

Even though inside the Model 4200-SCS the PMU LO terminal is connected to the GNDU (ground unit) common, it is not recommended to use the GNDU as the LO terminal of the PMU. Using the GNDU as the LO terminal of the PMU will limit the bandwidth of the circuit. This is because of the large inductive loop area that will be created when the HI and LO leads are separated in this way.

Minimize Cable Length

As a general rule for pulsing, use the shortest possible cable length to achieve the highest frequency output. Longer cable lengths have longer reflection times which can slow down the transmission times since it takes longer for the pulse to settle. Longer cables also have higher capacitance and may have impedance mismatches causing distortions such as multiple reflections in the waveform. Higher capacitance in the cabling causes higher capacitive charging effects during the pulse transition.

Connections to Chuck

If one of the device terminals is on the backside of the wafer, it is best to pulse only on that terminal (on chuck) and measure at another terminal (on the top side of the wafer) using the second channel. Try not to measure from the PMU channel that is connected to the chuck. This is similar to how CV measurements are made using a chuck.

When making PMU connections to the back side of a wafer, the user will give up some functionality of the PMU: fast transitions, high frequency, low current, etc. This is because the output of the high frequency PMU channel is connected to the chuck capacitance and the chuck cabling, which slows down the source response and couples noise into the measurement. If possible, it is better to use a manipulator to probe directly to the check. However, proceed with caution and verify the waveforms.

For high speed sourcing and measuring, it is best to use all top-side connections and avoid connecting the PMU to the chuck.

Summary of Proper Connections:

Using proper cabling and connections is important for ultra-fast I-V applications in order to achieve the highest frequency output and to avoid signal distortions and capacitive charging effects. Here is a summary of the proper connections:

- Use cabling and connections optimized for high frequency (at least 150 MHz)
- Use a signal path that matches the impedance of the instrument (50 ohm)
- Tie the low side of the DUT to the shield of the coax cable
- Connect the shields from each PMU channel together as close as possible to the DUT
- Minimize the loop area once the center conductor and shield are separated in the test circuit
- Minimize cable length